

GATE- ALL BRANCHES ENGINEERING MATHEMATICS



Linear Algebra

Lecture No. 3 ✓



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[Recap of previous lecture]



→ Orthogonal Matrix → $A^T A = A A^T = I$.

→ Involutary → $A^2 = I$

→ Idempotent → $A^2 = A$.

→ Nilpotent Matrix → $A^n = 0$ ($A \neq 0$)

→ Multiplication of Matrices

Even if $A \neq 0$ & $B \neq 0$

Still AB can be a Null Matrix.

Topics to be Covered



Topic

✓
Determinant of a Matrix

Topic

✓
Properties of Determinants

Topic

✓
Rank of a Matrix

Determinant of a matrix

→ Scaling factors (For Areas in 2D)
(For Volumes in 3D).

The Sum of Product of elements of a Row/column with its corresponding Co-factors. ; For a Matrix 'A' determinant is denoted as $|A|$ (or) $\det(A)$.

Let $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}_{3 \times 3}$

Minor of element a_{ij} :

$$\text{Minor of } a_{11} = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} = a_{22}a_{33} - a_{23}a_{32}$$

$$\text{Minor of } a_{12} = \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} = a_{21}a_{33} - a_{23}a_{31}$$

$$\text{Minor of } a_{13} = a_{21}a_{32} - a_{31}a_{22}$$

Co-factor of a_{ij} :

$$\text{Co-factor of } a_{ij} = (-1)^{i+j} \cdot \text{Minor of } a_{ij}$$

$$\rightarrow \text{Co-factor of } a_{11} = (-1)^{1+1} \cdot (a_{22}a_{33} - a_{23}a_{32}) = A_{11}$$

$$\text{Co-factor of } a_{12} = (-1)^{1+2} \cdot (a_{21}a_{33} - a_{23}a_{31}) = A_{12}$$

$$\text{Co-factor of } a_{13} = (-1)^{1+3} \cdot (a_{21}a_{32} - a_{31}a_{22}) = A_{13}$$

$$\therefore \text{considering } a_{11}A_{11} + a_{12}A_{12} + a_{13}A_{13}$$

$$= a_{11}(a_{22}a_{33} - a_{23}a_{32}) + (-1)a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{31}a_{22})$$

④ → The sum of product of elements of a row/column with co-factors of other row/column is always Zero.

Ex: Consider. $a_{11}A_{21} + a_{12}A_{22} + a_{13}A_{23}$.

$$= a_{11} \cdot (-1)^{2+1} \cdot (a_{12}a_{33} - a_{32}a_{13}) + a_{12} \cdot (-1)^{2+2} \cdot (a_{11}a_{33} - a_{13}a_{31}) + a_{13} \cdot (-1)^{2+3} \cdot (a_{11}a_{32} - a_{31}a_{12})$$

$$= -a_{11}a_{12}a_{33} + a_{11}a_{32}a_{13} + a_{12}a_{11}a_{33} - a_{12}a_{13}a_{31} - a_{13}a_{11}a_{32} + a_{13}a_{31}a_{12}$$

$$= 0$$

Co-factor Matrix: The Matrix formed by all the cofactors of the elements of a given Matrix.

i.e For a Matrix $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$; the co-factor Matrix is

$$\begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix}.$$

We consider Transpose of the co-factor Matrix. It's called "Adjoint Matrix".
It is denoted by $\text{adj}(A)$.

$$\therefore \text{adj}(A) = (\text{co-factor Matrix})^T.$$

$$\Rightarrow \text{adj}(A) = \begin{bmatrix} A_{11} & A_{21} & A_{31} \\ A_{12} & A_{22} & A_{32} \\ A_{13} & A_{23} & A_{33} \end{bmatrix}$$

$$\therefore \text{Considering } A \cdot \text{adj}(A) = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} A_{11} & A_{21} & A_{31} \\ A_{12} & A_{22} & A_{32} \\ A_{13} & A_{23} & A_{33} \end{bmatrix}$$

$$= \begin{bmatrix} |A| & 0 & 0 \\ 0 & |A| & 0 \\ 0 & 0 & |A| \end{bmatrix} = |A| \cdot \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = |A| \cdot I$$

⊗

$$\Rightarrow A \cdot \text{adj}(A) = |A| \cdot I$$

Multiplying A^{-1} on both sides $\Rightarrow (\cancel{A^{-1} \cdot A}) \text{adj}(A) = \cancel{A^{-1}} \cdot (|A| I)$

$$\Rightarrow \text{adj}(A) = |A| \cdot A^{-1} \Rightarrow A^{-1} = \frac{1}{|A|} \cdot \text{adj}(A) \text{ provided } |A| \neq 0.$$

Singular Matrix: A matrix 'A' is said to be singular if its determinant is zero; i.e. $\det(A) = 0$.



For 'A' to have a Multiplicative inverse; (i.e. For A^{-1} to exist), the Matrix 'A' must be Non-singular (i.e. $\det(A) \neq 0$).

→ $\det \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} =$

$= (4) + (5) + (6) - \{(1) + (2) + (3)\}.$

④ → Number of terms in the expansion of a ' $A_{n \times n}$ ' determinant is $n!$. 

For $n=3$; We have $3! = 6$ terms.

For $n=5$; We have $5! = 120$ terms.

Factorial of ' n '.

→ For $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix} = \{a_{11}(6 \text{ terms})\} - a_{12}(6 \text{ terms}) + a_{13}(6 \text{ terms}) - a_{14}(6 \text{ terms})$
 $= 24 \text{ terms} = 4! /$

Principle of Finite Mathematical Induction.



Properties of a determinant

For a Matrix $A_{n \times n}$;

- (i) If two rows/two columns are interchanged, then the determinant changes its sign. (Absolute Value remains the same).

$$\text{i.e. } \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = - \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{31} & a_{32} & a_{33} \\ a_{21} & a_{22} & a_{23} \end{vmatrix}.$$

(ii) The determinant of the Matrix remains unchanged by interchanging its rows & columns.

i.e. $\det(A) = \det(A^T)$

(iii) For $\alpha_1, \alpha_2, \alpha_3$ being Scalars,

$$\begin{vmatrix} \alpha_1 a_{11} & \alpha_2 a_{12} & \alpha_3 a_{13} \\ \alpha_1 a_{21} & \alpha_2 a_{22} & \alpha_3 a_{23} \\ \alpha_1 a_{31} & \alpha_2 a_{32} & \alpha_3 a_{33} \end{vmatrix} = \alpha_1 \cdot \alpha_2 \cdot \alpha_3 \cdot \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}.$$

~~(iv)~~ (iv) If 'k' is a Scalar, then $\det(kA_{n \times n}) = k^n \cdot \det(A_{n \times n})$

② The determinant of an upper triangular / a lower triangular / a diagonal matrix is product of principal diagonal elements of that matrix.

③ If the sum of all the elements of row/column is zero for all rows/columns, then the determinant is zero.

i.e For $A = \begin{bmatrix} a_{11} + a_{12} + a_{13} \\ a_{21} + a_{22} + a_{23} \\ a_{31} + a_{32} + a_{33} \end{bmatrix}$; if $\left. \begin{array}{l} a_{11} + a_{12} + a_{13} = 0 \\ a_{21} + a_{22} + a_{23} = 0 \\ a_{31} + a_{32} + a_{33} = 0 \end{array} \right\} \Rightarrow |A| = 0.$

④ The determinant of a Skew-symmetric Matrix of odd order is zero. (i.e. If ' $A_{n \times n}$ ' is a skew-symmetric Matrix;

then $\det(A) = 0$ if ' n ' is odd).

Proof:

If $A = -A^T \rightarrow$ skew-symmetric

$$\begin{aligned} \text{then } \det(A) &= \det(-A^T) \\ &= (-1)^n \cdot \det(A^T) \end{aligned}$$

$$\Rightarrow \det(A) = (-1)^n \cdot \det(A)$$

$$\Rightarrow \{1 - (-1)^n\} \det(A) = 0$$

For ' n ' being odd;

$$1 - (-1)^n = 2$$

$$\Rightarrow 2 \cdot \det(A) = 0$$

$$\Rightarrow \boxed{\det(A) = 0}$$

⑤ The determinant of an orthogonal Matrix is always ± 1 .



Consider $A \cdot A^T = I$.

$$\Rightarrow \det(A \cdot A^T) = \det(I)$$

$$\Rightarrow |A| \cdot |A^T| = 1.$$

$$(\because |A| = |A^T|)$$

$$(\det(ABCD) = |A| \cdot |B| \cdot |C| \cdot |D|).$$

$$\Rightarrow |A|^{\checkmark} = 1 \Rightarrow \boxed{|A| = \pm 1}$$

$\Rightarrow |A|$ is Never 0 for an orthogonal Matrix 'A'

$\Rightarrow A^{-1}$ always exist & $\boxed{A^{-1} = A^T}$

⑥ The determinant of an Involutary Matrix is ± 1 .



$$A^{\check{}} = I.$$

$$\Rightarrow \det(A^{\check{}}) = \det(I); \quad (\because |A^n| = |A|^n)$$

$$\Rightarrow |A|^{\check{}} = 1$$

$$\Rightarrow |A| = \pm 1$$

$$A^{-1} = A$$

$$\begin{aligned} |A^{\check{}}| &= |A \cdot A| \\ &= |A| \cdot |A| = |A|^{\check{}} \end{aligned}$$

⑦ The determinant of an Idempotent Matrix is 0 (or) 1.

$$A^{\check{}} = A$$

$$\Rightarrow \det(A^{\check{}}) = \det(A) \Rightarrow |A|^{\check{}} = |A| \Rightarrow |A| \{ \underbrace{|A| - 1} \} = 0$$

$$\Rightarrow |A| = 0 \text{ (or) } |A| = 1$$

⑧ If Product of two Non-zero Matrices 'A' & 'B' (i.e. say AB) is equal to zero, then Both 'A' & 'B' must be Singular Matrices.

Proof:

$$AB=0 \Rightarrow \det(AB)=0 \Rightarrow \underline{|A| \cdot |B|=0}$$

Case-1: Let $|A|=0$ & $|B| \neq 0$.

$\Rightarrow B^{-1}$ Exist.

$$\text{Given } AB=0 \Rightarrow A(B B^{-1}) = 0 \cdot B^{-1}$$

$\Rightarrow A=0 \rightarrow \text{False. (since } A \neq 0; B \neq 0)$

Case-2: Let $|A| \neq 0$ & $|B|=0$.

$$\Rightarrow A^{-1} \text{ Exist} \Rightarrow A^{-1}(AB) = A^{-1} \cdot 0 \Rightarrow B=0 \rightarrow \text{False.}$$

→ If 'A' is a '2x2' real symmetric Matrix with trace '12'; then the Maximum Possible Value of determinant of 'A' is 36.

Sol:

Let $A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$; Given $\text{tr}(A) = 12$
 $\Rightarrow a + c = 12$.

$\det(A) = ac - b^2 \Rightarrow$ For $a, b, c \in \mathbb{R}$; $\text{Max}(ac - b^2) = \text{Max}(ac)$.
 $b^2 \geq 0$

$\Rightarrow$ For $ac - b^2$ to be Maximum; $b^2 = 0$

$\Rightarrow$ Maximum of ac when $a + c = 12$
 $\Rightarrow c = 12 - a$

$\Rightarrow ac = a(12 - a) = 12a - a^2$

$\left. \begin{array}{l} \text{Maximum of } ac \text{ when } a + c = 12 \\ \Rightarrow c = 12 - a \end{array} \right\} \begin{array}{l} \frac{a+c}{2} \geq \sqrt{ac} \Rightarrow \frac{12}{2} \geq \sqrt{ac} \\ \Rightarrow ac \leq 36 \end{array}$

$\therefore$ Max. of $12a - a^2 \Rightarrow \frac{d}{da}(12a - a^2) = 0 \Rightarrow a = 6 \Rightarrow c = 6 \Rightarrow \text{Max}(ac) = 36$

→ Determinant of adj(A):

For a Matrix $A_{n \times n}$

$$\det(kI) \propto^n \det(I)$$

$$A \cdot \text{adj}(A) = |A| \cdot I$$

$$\Rightarrow \det(A \cdot \text{adj}(A)) = \det(|A| I)$$

$$\Rightarrow \cancel{\det(A)} \cdot \det(\text{adj}(A)) = |A|^n \cdot \cancel{\det(I)}^{(n-1)}$$

$$\Rightarrow \boxed{\det(\text{adj}(A)) = |A|^{n-1}}$$

→ $\det(\text{adj}(\text{adj} A)) = ?$

$$\text{Let } \text{adj} A = B \Rightarrow |B| = |\text{adj} A| = |A|^{n-1}$$

$$\Rightarrow B \cdot \text{adj}(B) = |B| \cdot I$$

$$\Rightarrow \det(\text{adj} B) = |B|^{n-1}$$

$$\Rightarrow \det(\text{adj}(\text{adj} A)) = (|A|^{n-1})^{(n-1)}$$

$$\Rightarrow \boxed{\det(\text{adj}(\text{adj} A)) = |A|^{(n-1)^2}}$$

$$|B| = |A| \quad \begin{matrix} B = -A \\ 2 \times 2 \end{matrix} \Rightarrow A+B=0$$

$$\underbrace{|A+B|}_{=0} \neq \underbrace{|A|+|B|}_{2|A|}$$



2 mins Summary



- Determinants.
- Properties of determinants.



THANK - YOU